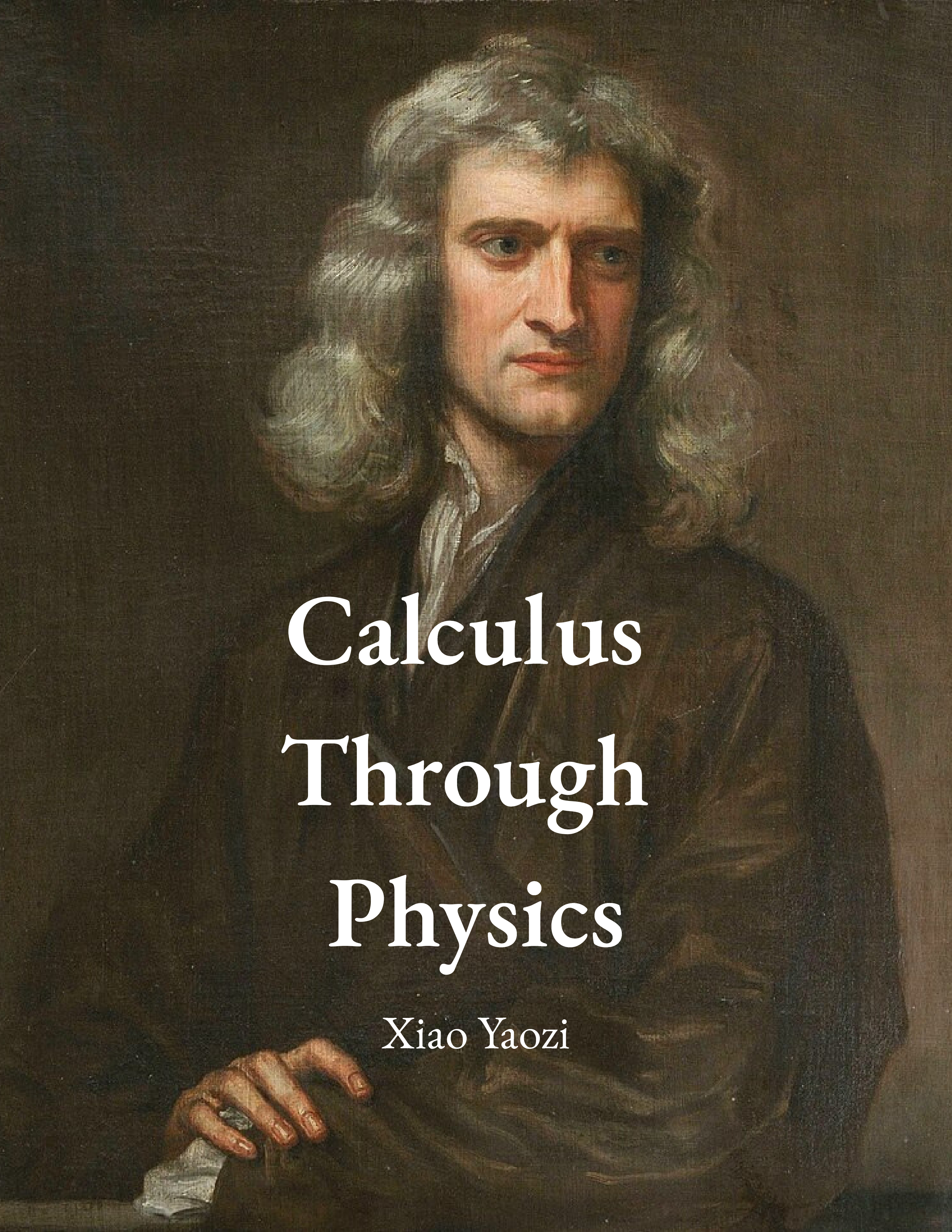

Calculus Through Physics

Xiao Yaozi

Contents

Preface I: Calculus, the language of the universe

As you are seated right here, on your seat, reading this book, you must realize that you are made of the same fundamental particles as everything in the universe. These fundamental particles are governed by the Schrodinger equation, which is written with notation from calculus [1]:

[1] *We will assume that you are not fast enough for relativistic effects to apply.*

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla\Psi \;\; V\Psi \tag{0.0.1}$$

Do not worry if you do not understand this equation. In fact, this book will teach some quantum mechanics at the end because it just shows the absolute manifestation of calculus in physics. Even if you know absolutely nothing about quantum mechanics, I have a much more subtle, yet interesting question for you: why is the area of a circle of radius r is πr^2. You might turn to the rearrangement proof, or the "pizza" explanation as I like to call it. Take a circle, and divide it into congruent segments like cutting pizza slices. Then, take the slices and rearrange them in such way such that they form a sort of parallelogram.

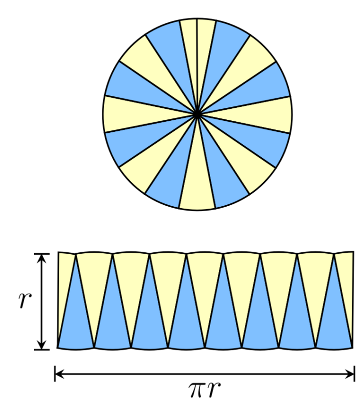

Figure 0.1 Figure of the rearrangement proof.

Something that you notice in Figure 0.1 is that if you make each slice smaller, the parallelogram will become more like a rectangle. This, right here, is a very subtle yet profound fact that makes many problems in the real world much easier. As the famous saying goes, "calculus turns one big problem into infinitely smaller ones." This little trick is the reason for developing the calculus method. Everything in calculus stems from the proposition that dividing something into infinitesimal parts reduces the complexity of a problem. So far, this has held up well. No matter if you are looking at classical mechanics or quantum mechanics, all the equations are written with calculus. While this little fact may seem very deep or even philosophical, and we will explore on the paradoxes of infinitesimals, I believe that one cannot meaningfully know what calculus means before knowing what it does. The purpose of this book is to teach you how to do calculus through the lens of physics. There are many schools of thought on the pedagogy of calculus and there are great texts on calculus such as Stewart, J. *Calculus: Early Transcentals, 8th Ed.* and Strang, G. *Calculus*. Furthermore, if you already know the basics of calculus, I would also recommend Arfken, G. *Mathematical Methods for Physicists* and Boas, M.L. *Mathematical Methods in the Physical Sciences* is also a classic. While these books are fantastic, they either already assume a decent knowledge of calculus or does not put its focus on physics. This book seeks to fill this special niche.

I wish every reader good luck on their journey on learning calculus. It is not trivial and rereading confusing parts of the book is highly recommended. More than just knowing how to apply calculus however, I also encourage the reader to appreciate the beauty of calculus and its unreasonable effectiveness in describing our physical world. To end off, thanks for choosing this book. Happy reading.

Xiao Yaozi

Preface II: How to read this book

This book is structured into four parts. The first part deals with precalculus or the content that is needed for the later calculus chapters. Be warned, however, that the first part is rather short and sweeps over the content very quickly. The second part corresponds to the standard curriculum of Calculus I and is the beginning of the actual calculus portion. Similarly, the third and fourth part corresponds to Calculus II and multivariable calculus respectively. As with any textbook, problems are included at the end of each chapter for the reader to practice and familiarise with the topic. To aid the reader, we have included the following notation for denoting the type of problem:

1. ***** means that the problem is an *essential* problem, usually something that is very good to know or a theorem not derived in the text.

2. ****** means that the problem is more *peripheral* and the solution might involve using a clever identity or trick.

3. ******* means that the problem is very *difficult* and might take the reader very long to solve.

4. Nothing means that the problem is not very important and can be skipped.

Furthermore, to prevent confusion on sidenotes[2] and indices, sidenote markers are **bolded** while the latter is not.

[2] *This is a sidenote. It is italicized to prevent confusion with the main text.*

Finally, if you have any questions or errata, please contact me and submit an issue on GitHub. The GitHub link is https://github.com/xyzdev-code/Calculus-through-Physics.

1.4 An aside: A more rigorous definition of a function

TBD

2.6 An aside: Bra-ket and Hilbert space

TBD

2.7 An aside: Eigenfunctions and eigenvalues

TBD

Part II

Calculus I

Chapter

3 Derivatives

3.1 Differentiation basics

Onto the calculus portion that is very much anticipated. No more the quick skimping over precalculus topics. We shall enter this discussion with a very simple question that will yield many interesting results. Take a look at the graph below which describes the displacement of particle x over time t [31]:

[31] *x starts at 0, because if I were to plot the graph with negative values of x, it does not make sense physically. Unfortunately, this makes the cubic nature of the graph less obvious.*

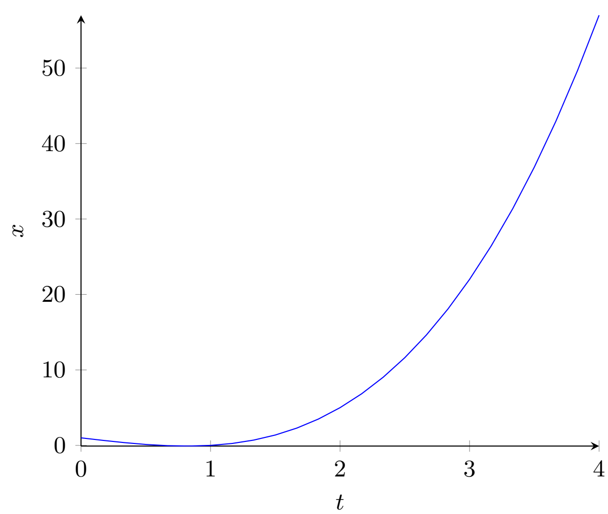

Figure 3.1 Graph of $f t = t^3 - 2t \;\; 1$.

From Figure 3.1, we see that this is a cubic graph. It should not to be hard to see that the graph is always changing value. This corresponds to the fact that the velocity of the particle is not constant. Also, it should not be difficult to see that the rate of change of the graph is not constant. This, meanwhile, shows that the acceleration of the particle is not constant. We can use the gradient formula to verify this:

Definition 3.1 Gradient

The gradient between two points x, fx and $x \;\; a, fx \;\; a$ is:

$$\textit{Gradient} = \frac{\Delta y}{\Delta x} = \frac{fx \;\; a - fx}{a} \tag{3.1.1}$$

In case you have forgotten, the rate of change of displacement with respect to time is velocity, and the rate of change of velocity with respect to time is acceleration. Hence, we can use the gradient formula to find the average velocity of the particle over a certain time interval. For example, if we want to find the average velocity between $t = 1$ and $t = 2$, we can use Equation (3.1.1):

$$\begin{aligned}\textit{Average velocity} &= \frac{f2 - f1}{2 - 1} \\ &= \frac{2^3 - 2 \cdot 2 \;\; 1 - 1^3 - 2 \cdot 1 \;\; 1}{1} \\ &= \frac{5}{1} \\ &= 5\end{aligned}$$

Using Equation (3.1.1) again, we can pick some points on the graph. You would notice that if you plug in different points then the gradient is different. The main, central question is: if the rate of change is not constant, then what is the rate of change at any single point along the function? This is where the gradient formula fails. Our current formula can only give us the average velocity for some time period, however, we want something that gives us the gradient of the graph at every single point along the graph. In physics, this is analogous to the velocity of the particle at every single point along its journey, or its *instantaneous velocity*. The forefathers of calculus were all brilliant and they noticed a special fact that applies to many functions. Let us zoom in to the graph where $t = 2$, we want to zoom in as closely as we can.

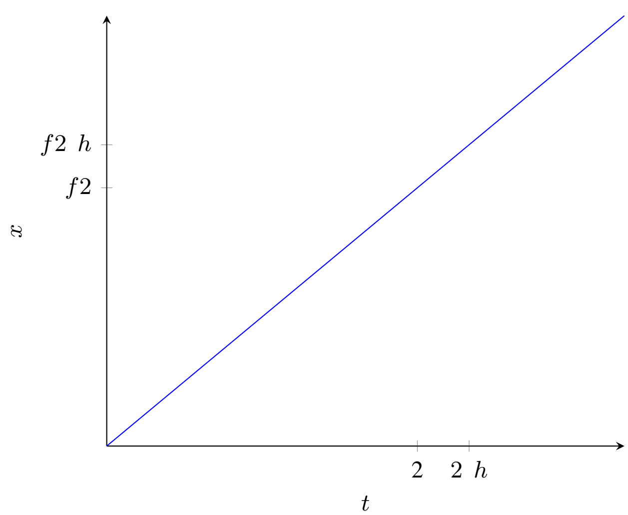

Figure 3.2 Zoomed in graph of Figure 3.1.

In Figure 3.2, we realize that the graph, when zoomed in sufficiently, essentially is a straight line. This is exciting, we can use our gradient formula which works on straight lines. In addition, on the x-axis, we mark $2 \ h$, a small distance away from 2. We can infer that h is a really tiny number, almost going to 0 as we keep zooming in. We use the limit notation to denote this. Hence, we can denote it as $\lim_{h\to 0}$. [32] You may be itching to use the gradient formula now. We can substitute in the values:

he limit notation and its concepts will be discussed in a later er, for now, just keep ind that this means h a really tiny number, almost 0.

$$\lim_{h\to 0} \frac{f 2 \ \ h - f 2}{h}$$

However, notice that we have to divide throughout by h, and h is very small, infinitesimally small. Some careless manipulation and you would say the expression blows up to infinity. What the formula does tell us, though, is the gradient practically at 2. This gives us an idea. Can we just replace 2 with any value of x? This brings us to the definition of the *derivative*.

Definition 3.2 Derivative

The *derivative* of a function $f x$ is the instantaneous rate of change of that function. It can be written as $f' x$, $\frac{df x}{dx}$ or f.[33]

$$f' x = \lim_{h\to 0} \frac{f x \ \ h - f x}{h} \tag{3.1.2}$$

$\star$

ometimes, to save g a few letters, the to the functions e omitted, such as f d of $f x$. This is more on when the function n multiple inputs.

In the definition of the derivative, we can see that the derivative is also a function. Also, the definition states that it is the "instantaneous rate of change". This simply means that we are looking at the rate of change of the function over a very, very tiny range of x. Of course, the definition may feel contradictory. How can some value change "instantaneously"? Well, this is resolved by not exactly by substituting $\Delta y = 0$, instead we let it "approach" 0. As a result, using this clever little mathematical trick, we avoided this problem. However, some of you may have noticed we have not evaluated any derivatives, we just defined them. It is no good to not answer our original question. To illustrate a derivation of a derivative, we shall use a much simpler graph, $f x = x^2$.

From Figure 3.3, we can consider what happens at $x = 2$. We shall use the definition of the derivatives, which is also more commonly referred to as "first principles".

Example 3.1 Derivative of $f x = x^2$

To find the derivative at $x = 2$, we use the first principles as such [34]:

$$\begin{aligned} f' 2 &= \lim_{h\to 0} \frac{f 2 \ \ h - f 2}{h} \\ &= \lim_{h\to 0} \frac{2 \ \ h^2 - 4}{h} \\ &= \lim_{h\to 0} \frac{4 \ \ 4h \ \ h^2 - 4}{h} \\ &= \lim_{h\to 0} 4 \ \ h \\ &= 4 \end{aligned}$$

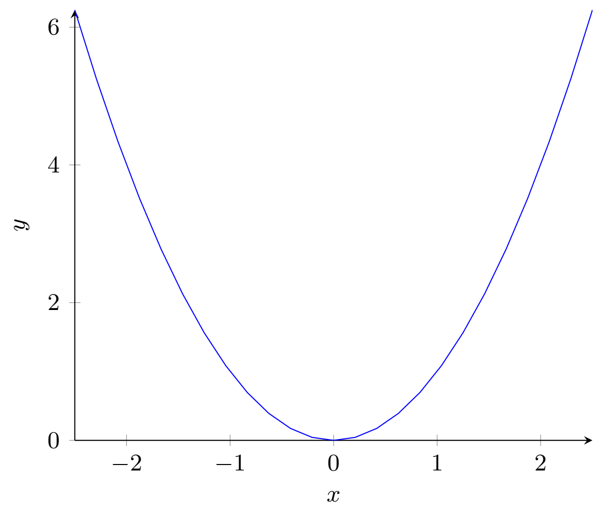

Figure 3.3 Figure of $y = x^2$.

To find the derivative of f, we use a similar manipulation:

$$\begin{aligned} f'x &= \lim_{h\to 0} \frac{fx\ \ h - fx}{h} \\ &= \lim_{h\to 0} \frac{x^2\ \ 2xh\ \ h^2 - x^2}{h} \\ &= \lim_{h\to 0} 2x\ \ h \\ &= 2x \end{aligned}$$

[34] *We can ignore h in the second last step because as a general rule of thumb, as long as substituting $h = 0$ does not create an expression of $\frac{a}{0}$, we can let $h = 0$. Of course, this is very not rigorous and there are other exceptions that will be very evident after the reader has read the later chapter of limits.*

In other words, we have also found the instantaneous rate of change of $fx = x^2$. We shall solve the problem we posed at the start of the chapter, what is the instantaneous rate of change or velocity for Figure 3.1.

Example 3.2 Derivative of $ft = t^3 - 2t\ \ 1$

Using the first principles, we have:

$$\begin{aligned} f't &= \lim_{h\to 0} \frac{ft\ \ h - ft}{h} \\ &= \lim_{h\to 0} \frac{t\ \ h^3 - 2t\ \ h\ \ 1 - t^3 - 2t\ \ 1}{h} \\ &= \lim_{h\to 0} \frac{t^3\ \ 3t^2h\ \ 3th^2\ \ h^3 - 2t - 2h\ \ 1 - t^3\ \ 2t - 1}{h} \\ &= \lim_{h\to 0} 3t^2\ \ 3th\ \ h^2 - 2 \end{aligned}$$

$$= 3t^2 - 2$$

Therefore, the instantaneous velocity of the particle at any time t is $3t^2 - 2$. For example, at $t = 1$, the instantaneous velocity is simply 1, and at $t = 2$, the instantaneous velocity is 10. Remember how we found the average velocity between $t = 1$ and $t = 2$ to be 5? Notice how the instantaneous velocity at $t = 1$ is less than 5, and at $t = 2$ is greater than 5. By looking at the graph, we can see that this makes sense. The value at $f1$ is smaller than the value of $f2$, hence the average velocity is greater than the instantaneous velocity at $t = 1$. Similarly, the instantaneous velocity at $t = 2$ is greater than the average velocity.

△

Then, we shall touch on something that is very similar to the derivative, the tangent line. The tangent line is a straight line that just touches the curve at one point. The gradient of the tangent line is equal to the gradient of the curve at that point. This means that if we know the derivative of a function at some point, we can find the equation of the tangent line at that point. Remember, the derivative is the instantaneous rate of change of the function. You will find the equation of the tangent line to be very useful in many instances. The exact derivation will be left to Problem 3.3, but given a function fx, the equation of the tangent line at $x = a$ is:

$$y = f'ax - a \ \ fa \tag{3.1.3}$$

Newton's method

Example 3.3 Newton's method

Newton's method is a method to approximate the roots of a function. It is based on the idea that if we have a guess x_0 that is close to the root, then we can use the tangent line at x_0 to find a better approximation x_1. The equation of the tangent line at x_0 is:

$$y = f'x_0x - x_0 \ \ fx_0 \tag{3.1.4}$$

To find the root of the tangent line, we set $y = 0$:

$$\begin{aligned} 0 &= f'x_0x - x_0 \ \ fx_0 \\ -fx_0 &= f'x_0x - x_0 \\ x - x_0 &= -\frac{fx_0}{f'x_0} \\ x &= x_0 - \frac{fx_0}{f'x_0} \end{aligned}$$

Therefore, we can use the formula:

$$x_{n1} = x_n - \frac{fx_n}{f'x_n} \tag{3.1.5}$$

We shall use the above to find a better approximation of the root. We can repeat this process until we reach a satisfactory level of accuracy. Take the function $fx = x^2 - x - 1$ as an example. We want to find the root of this function. We can start with an initial guess of $x_0 = 1$. The derivative of the function is $f'x = 2x - 1$. We can use Newton's method to find a better approximation:

$$\begin{aligned} x_1 &= x_0 - \frac{fx_0}{f'x_0} \\ &= 1 - \frac{1^2 - 1 - 1}{2 \cdot 1 - 1} \\ &= 2 \end{aligned}$$

Repeating the process:

$$\begin{aligned} x_2 &= x_1 - \frac{fx_1}{f'x_1} \\ &= 2 - \frac{2^2 - 2 - 1}{2 \cdot 2 - 1} \\ &= \frac{5}{3} \approx 1.6667 \end{aligned}$$

Continuing:

$$\begin{aligned} x_3 &= x_2 - \frac{fx_2}{f'x_2} \\ &= \frac{5}{3} - \frac{\left(\frac{5}{3}\right)^2 - \frac{5}{3} - 1}{2 \cdot \frac{5}{3} - 1} \\ &= \frac{13}{8} = 1.625 \end{aligned}$$

And again (finally):

$$\begin{aligned} x_4 &= x_3 - \frac{fx_3}{f'x_3} \\ &= \frac{13}{8} - \frac{\left(\frac{13}{8}\right)^2 - \frac{13}{8} - 1}{2 \cdot \frac{13}{8} - 1} \\ &= \frac{21}{13} \approx 1.6154 \end{aligned}$$

If you were wondering what is the exact solution to the polynomial, it is in fact the golden ratio ϕ, which is about 1.618, very close to our fourth iteration (about 99.8% accurate).

Lastly, I would like to comment that normally, if not explicitly stated in the question to use "first principles", you can use the derivative rules you will learn in the next chapter to solve derivatives much more quickly and efficiently. In all honesty, using the "first principles" is very tedious and generally would not be used.

*** Problem 3.1**

Using first principles, find the derivatives of the following functions:

a) $fx = x^2 \ \ 1$

b) $gx = x^3 \ \ x$

c) $hx = \frac{1}{x}$

*** Problem 3.2**

Let $fx = x - 1x - 2 \cdots x - 1729$, find $f'1729$. *(Adapted from SMO 2025 Senior Section Round 1).*

*** Problem 3.3**

Given a differentiable function fx, find the equation of the tangent line at $x = a$.